

Finally, **escape learning** or **active avoidance learning** (lower right) occurs if the *responses lead to escape from or avoidance of something painful*. One therefore learns to make the response, even though it is followed by nothing happening. Examples are learning to come indoors when a storm is brewing (to avoid getting wet) and learning to get out of the way when you hear a driver blowing a horn (to avoid getting hit). Because the *response is reinforced by the absence of a painful event*, escape or avoidance learning is also sometimes known as *negative reinforcement*. Note that a negative reinforcement is not a punishment; reinforcement increases a behavior. The difference between positive and negative reinforcement is whether the reinforcement is the presentation of something (like food) or the avoidance of something (like pain). The terms “escape learning” and “avoidance learning” are more widely used and less confusing than the term “negative reinforcement” (see Kimble, 1993), but you should understand all of these terms.

In summary, responses are increased by gaining something like food (positive reinforcement) or avoiding something like pain (escape or avoidance training or negative reinforcement). Responses are decreased by receiving pain (punishment) or by losing a chance for food or the like (omission training).

CONCEPT CHECK

8. Identify each of the following examples, using the terms shown in Table 6.2:
 - a. Your employer gives you bonus pay for working overtime.
 - b. You learn to stop playing your accordion at 5:00 A.M., because your roommate threatens to kill you if you do it again.
 - c. You turn off a dripping faucet, ending the “drip drip” sound.
 - d. You drink less beer than you once did, because you feel sick after drinking more than one glass.
 - e. Your swimming coach says you cannot go to the next swim meet (which you were looking forward to), because you broke a training rule.
 - f. Because you drive recklessly, you temporarily lose the privilege of driving the family car.
 - g. You stay away from some fellow students who are coughing and sneezing, because you do not want to catch whatever illness they have. (Check your answers on page 218.)

How effective is punishment? That depends. If an individual has only one way to satisfy a strong motivation, punishing that act is generally ineffective. In one experiment, B. F. Skinner (1938) first trained food-deprived rats to press a bar to get food and then stopped delivering food for bar presses—an extinction procedure. For the first 10 minutes of extinction, some rats not only failed to get food

but the bar slapped their paws every time they tried to press it. They temporarily suppressed their bar-pressing, but in the long run they made as many presses as did the rats that never received any punishment.

Skinner concluded that punishment was ineffective, except for a temporary suppression of behavior. That conclusion, however, is an overstatement from the results of a single experiment (Staddon, 1993). Punishment can be quite effective, if it occurs quickly and consistently after the punished response and if alternative responses are available (Walters & Grusec, 1977). For example, if you want to teach your 3-year-old daughter not to put her hand on the stove, a gentle slap on the hand, or even a stern, sharp “NO!” is more effective than reinforcing her for another, competing behavior. Note that the quickness and consistency of the rebuke are important; making the punishment more intense will not make it more effective. Children respond more readily to mild punishment accompanied by a parent’s explanation of why they were punished than they do to more intense punishment without an explanation.

Punishment is least effective when it is applied in response to a strongly motivated behavior for which the individual has no alternative. For example, if you were punished for eating or drinking, you would continue to do so anyway. In the experiment in which Skinner reported punishment to be ineffective, he was punishing food-deprived rats for pressing the lever that had in the past produced food. Bar-pressing was the only food-getting behavior the rats could perform. Punishment can also be ineffective with attention-starved children who find that they can get more attention for “bad” behavior than for “good” behavior.



Being ejected from a game for using foul language is an example of effective punishment: It is quick and consistent, and the player has other available behaviors that can achieve good results without incurring punishment.

And punishment can backfire in other ways. For example, a parent who spansks a child for nervous fidgeting may find that the spanking makes the child even more nervous and fidgety. In short, punishment can be either effective or ineffective, depending on a variety of circumstances.

CONCEPT CHECK

9. The U.S. government imposes strict punishments for selling illegal drugs. Based on what you have just read, why are those punishments ineffective for many people? (Check your answer on page 218.)

SOMETHING TO THINK ABOUT

Your local school board proposes to improve class attendance by lowering the grades of any student who misses a certain number of classes. Might the board achieve the same goal more effectively by using positive reinforcement? ★

What Constitutes Reinforcement?

With operant conditioning, a response becomes more common because it is followed by a reinforcer. But what is a reinforcer? It is “something that increases the probability of the preceding response.” So far, we are just going around in circles. It would be helpful to specify the basis of reinforcement, so we would know better how to reinforce a child for studying or how to reinforce a worker for doing a job well.

Thorndike suggested a trial-and-error approach to finding reinforcers: We could simply try a number of likely reinforcers to see what works. If some event serves as a reinforcer for one behavior, it will also serve as a reinforcer for other behaviors. That approach is, however, theoretically unsatisfactory. Why is food reinforcing, for example?

Many reinforcers satisfy biological needs (hunger, thirst, sex, temperature regulation, and so forth). However, for many people, alcohol and tobacco, which can be biologically harmful, act as stronger reinforcers than exercise and a healthy diet. Saccharin, a sweet but biologically useless chemical, can also be a reinforcer.

David Premack (1965) proposed a simple rule: *The opportunity to engage in frequent behavior (such as eating) will be a reinforcer for any less-frequent behavior (such as lever-pressing).* This relationship is known as the **Premack principle**. Thus, for example, if you spend more time reading than watching television, someone could increase your television watching by reinforcing you with books. For someone else, it might be possible to reinforce reading with opportunities to watch television. In short, a given opportunity may or may not be a reinforcer, depending on the individual and the response to be reinforced.

The limitation of the Premack principle is that we are sometimes reinforced by an opportunity to do an uncommon behavior. For example, you probably spend little time clipping



Different items can serve as reinforcements for different people. This man has devoted an enormous amount of time, effort, and money to collecting old comic books. Some people pay thousands of dollars for old stamps, coins, or baseball cards.

your toenails, but if you are badly overdue for clipping them, the opportunity to do so can be a reinforcer. According to the **disequilibrium principle** of reinforcement, *each of us has a normal, or “equilibrium,” state in which we divide our time among various activities in some preferred way, and if we are removed from that state, a return to it will be reinforcing.* If you have recently spent less time than usual clipping toenails, playing video games, or talking with friends, then the opportunity to do so will be a reinforcer. However, if you have already completed your usual amount of an activity, then the opportunity to do it again will be an ineffective reinforcer (Timberlake & Farmer-Dougan, 1991).

CONCEPT CHECK

10. Anorexia nervosa is a condition in which someone is obsessed with weight loss and refuses to eat enough to survive. Suppose you are trying to encourage such a person to eat and you wish to use positive reinforcement. According to the disequilibrium principle, how should you begin? (Check your answer on page 218.)

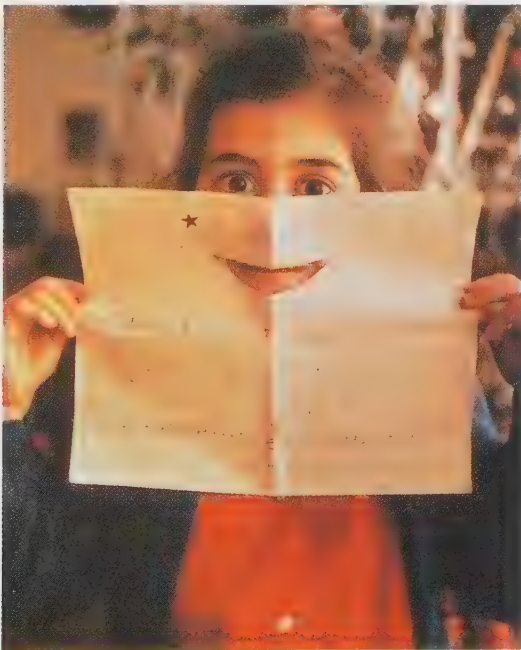
Unconditioned and Conditioned Reinforcement Food and drink are reinforcers for virtually everyone, even infants. By contrast, money is a reinforcer only for those who have learned what they can do with it. Psychologists therefore distinguish between **unconditioned reinforcers, which are reinforcing because of their own properties**, and **conditioned reinforcers, which became reinforcing because of their previous association with an unconditioned reinforcer**. Food and water are unconditioned reinforcers. Coins and bills (conditioned reinforcers) have no value in themselves, but become reinforcing because they can be exchanged for food or other

unconditioned reinforcers. A student learns that good grades will win the approval of parents and teachers; an employee learns that increased sales will win the approval of an employer. We spend most of our time working for conditioned reinforcers.

Reinforcement as Learning What Leads to What Thorndike, you will recall, believed that reinforcement strengthens the response that preceded it. According to that view, reinforcement is a mechanical process; the animal or person who experiences reinforcement simply engages in the response more frequently, without necessarily understanding *why*.

According to another view, individuals learn what leads to what (Tolman, 1932). A rat may learn that running down an alley leads to food. Having learned that, the rat does not automatically go running down the alley all the time. It runs down the alley only when it needs food.

For example, suppose an animal has managed to find its way through a maze. We present it with something it does not need at the moment, such as food just after it has eaten a meal. If we test the rat again after its next meal, it shows no increased speed at getting through the maze. However, if tested several hours after its most recent meal, the rat scurries through the maze, indicating that it had learned more than it had previously shown (Tolman & Honzik, 1930). This procedure is known as *latent learning*, because what is learned is at first *latent* (not apparent to the observer). Latent learning is generally taken as evidence that operant conditioning establishes cognitions or knowledge and does more than just strengthen response tendencies.



Many conditioned reinforcers are surprisingly powerful. Consider, for example, how hard some first-graders will work for a little gold star that the teacher pastes on an assignment.

Another example: A rat is reinforced with sugar water for making response A and is reinforced with food pellets for making response B. After both responses have become well established, the rat is made ill after consuming, say, the sugar water. Now, when given the opportunity to make either response, the rat makes mostly response B. Evidently, the original training did not just increase two responses; the rat learned which response produced which outcome (Colwill, 1993).

In other words, we must distinguish between learning and performance. A subject's behavior depends on which outcome is associated with that behavior and how strongly motivated the subject is to achieve the outcome.

Schedules of Reinforcement

The simplest procedure in operant conditioning is to provide reinforcement every time the correct response occurs. **Continuous reinforcement** refers to *reinforcement for every correct response*. As you know, not every response in the real world leads to reinforcement.

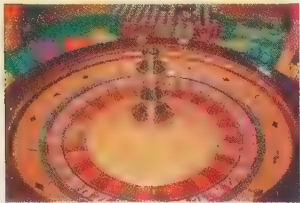
Reinforcement for some responses and not for others is known as **intermittent reinforcement**. We behave differently when we know that only some of our responses will be reinforced. Psychologists have investigated the effects of many **schedules of reinforcement**, which are *rules or procedures for the delivery of reinforcement*. Four schedules for delivery of intermittent reinforcement are fixed ratio, fixed interval, variable ratio, and variable interval. (See Table 6.3.) A ratio schedule provides reinforcements depending on the number of responses. An interval schedule provides reinforcements depending on the timing of responses.

Fixed-Ratio Schedule A fixed-ratio schedule provides a reinforcement only after a certain (fixed) number of correct responses have been made—after every fifth response, for example. Some animals continue responding even with a fixed-ratio schedule that reinforces every one-hundredth or every two-hundredth response. We see similar behavior among pieceworkers in a factory, whose pay depends on how many pieces they turn out or among fruit pickers who get paid by the bushel.

The response rate for a fixed-ratio schedule tends to be rapid and steady. However, if the schedule requires a large number of responses for reinforcement, there may be a temporary interruption after each reinforced response. For example, if you have just completed 10 calculus problems, you may pause briefly before starting your French assignment; after completing 100 problems, you would pause even longer.

Variable-Ratio Schedule A variable-ratio schedule is similar to a fixed-ratio schedule except that *reinforcement is provided after a variable number of correct responses*. For example, reinforcement may come after an average of 10 responses, but sometimes comes after as

TABLE 6.3 Some Schedules of Reinforcement

	TYPE	DESCRIPTION	
	Continuous	Reinforcement for every response of the correct type	
	Fixed ratio	Reinforcement following completion of a specific number of responses	
	Variable ratio	Reinforcement for an unpredictable number of responses that varies around a mean value	
	Fixed interval	Reinforcement for the first response that follows a given delay since the previous reinforcement	
	Variable interval	Reinforcement for the first response that follows an unpredictable delay (varying around a mean value) since the previous reinforcement	

few as 1 or 2 and sometimes after 20 or more responses. Variable-ratio schedules generate steady response rates. Gambling is reinforced on a variable-ratio schedule, because the gambler receives payment on an irregular basis for some responses and not for others.

In everyday life, we sometimes perform a behavior in which every response has an equal, fairly low probability of yielding a reinforcement. When that is true, we are reinforced on a variable-ratio schedule. For example, every time you apply for a job, you might or might not get it. The more times you apply, the better your chances, but you cannot predict how many applications you need to send in before you get a job offer.

Fixed-Interval Schedule A fixed-interval schedule provides reinforcement for the first response made after a specific time interval. For instance, an animal might get food for only the first response it makes after each 2-minute interval. Then it would have to wait another 2 minutes before another response would be effective. Animals (including humans) on such a schedule usually learn to pause after each reinforcement and begin to respond again only as the end of the time interval approaches.

Checking your mailbox is an example of behavior on a fixed-interval schedule. If your mail is delivered at about 3:00 P.M., you will get no reinforcement for checking your mailbox at 2:00 P.M. If you are eagerly awaiting an important letter, you might begin to check around 2:30 and continue checking every few minutes until it arrives.

Variable-Interval Schedule In a variable-interval schedule, reinforcement is available after a variable amount of time has elapsed. For example, reinforcement may come for the first response after 2 minutes, then for the first response after 7 seconds, then for the first response after 3 minutes 20 seconds, and so forth. There is no way to know how much time will pass before the next response is reinforced. Consequently, responses on a variable-interval schedule occur at a slow but steady rate. For example, you call your best friend and you get a busy signal. You don't

know how soon your friend will hang up, so you try again every few minutes.

Stargazing is another example of a response that is reinforced on a variable-interval schedule. The reinforcement for stargazing—seeing a new comet, for example—appears at irregular, unpredictable intervals. Consequently, both professional and amateur astronomers scan the skies regularly.

Extinction of Responses Maintained by Ratio or Interval Reinforcement After a schedule of intermittent reinforcement (either a ratio schedule or an interval schedule), the extinction of responses tends to be slower than it is after a schedule of continuous reinforcement (reinforcement for every response). For example, suppose that every time you have put money into a vending machine you received a can of soda (continuous reinforcement), but now you put in your money and the machine gives you nothing. You will not try many times before you quit. However, suppose you have frequently put money into a slot machine (a gambling device that reinforces on a variable-ratio schedule), sometimes winning some money but usually not. When the machine no longer produces reinforcements (ever), you might continue putting money into it for quite a while before your responses extinguish.

CONCEPT CHECKS

- Identify which schedule of reinforcement applies to each of the following examples:
 - You attend every new movie that appears at your local theater; you find most of them dull (not reinforcing) but really enjoy about one-fourth of them.
 - You occasionally check your e-mail to find out whether you have any new messages.
 - You tune your television set to an all-news cable channel, and you look up from your studies to check the sports scores every 30 minutes.
- Stargazing in the hope of finding a comet was cited as an example of a variable-interval schedule. Why is it not an example of a variable ratio?

- 13.** A novice gambler and a longtime gambler both lose 20 bets in a row. Which one is more likely to continue betting? Why? (Check your answers on page 218.)

Practical Applications of Operant Conditioning

Although operant conditioning arose from purely theoretical concerns, it has a long history of practical applications. Here are four examples.

Animal Training

Most animal acts today are based on training methods similar to Skinner's. To induce an animal to perform a trick, the trainer first trains it to perform a simple act that is similar to its natural behavior. Then the trainer shapes the animal, step by step, to perform progressively more complex behaviors. Most animal trainers rely on positive reinforcement rather than on punishment.

During World War II, Skinner proposed a military application of his training methods (Skinner, 1960). The military was having trouble designing a guidance system for its air-to-ground missiles. It needed apparatus that could recognize a target and guide a missile toward it but that was compact enough to leave room for explosives. Skinner said that he could teach pigeons to recognize a target and peck in its direction. If pigeons were placed in the nose cone of a missile, the direction of their pecking would guide the missile to the target. Skinner demonstrated that pigeons would do the job more cheaply and

more accurately and would take up less space than the apparatus then in use, but the military laughed off the whole idea.

Persuasion

How could you persuade someone to do something that he or she did not want to do? To use an extreme example, how could you convince a prisoner of war to cooperate with the enemy?

The best way is to start by reinforcing a very small degree of cooperation and then working up to the goal, little by little. This principle has been applied by people who had probably never heard of B. F. Skinner, positive reinforcement, or shaping. During the Korean War, the Chinese Communists forwarded some of the letters written home by prisoners of war but intercepted others. (The prisoners could tell from the replies which letters had been forwarded.) The prisoners began to suspect that they would have better luck getting their letters through if they said something mildly favorable about their captors. So, from time to time, they would include a brief remark that the Communists were not really so bad, or that certain aspects of the Chinese system seemed to work pretty well, or that they hoped the war would end soon.

After a while, the Chinese captors devised essay contests; the soldier who wrote the best essay (in the captors' opinion) would win a little extra food or some other privilege. Most of the winning essays contained a statement or two that complimented the Communists on a minor matter or admitted that "the United States is not perfect."

Gradually, more and more soldiers started to include such statements in their essays. Occasionally, the Chinese might ask one of them, "You said the United States is not



The high-tech hope of robots handling housekeeping chores has yet to materialize, but in the meantime, simian aides—trained monkeys—are helping the disabled. Monkeys are proving useful for indoor tasks for people with limited mobility. The monkey at left is being trained to retrieve objects identified with a laser beam. Such training relies on shaping behavior Skinner-style—building a new response by reinforcing sequential approximations to it.

perfect. We wonder whether you could tell us some of the ways in which it is not perfect, so that we can better understand your system.” Then they would ask the soldiers who had cooperated to read aloud their lists of what was wrong with the United States. And so on. Gradually, without torture or coercion and with only modest reinforcements, the Chinese induced many prisoners to make public statements denouncing the United States, to make false confessions, to inform on fellow prisoners, and even to reveal military secrets (Cialdini, 1993).

The point is clear: Whether we want to get rats to salute the flag or soldiers to denounce it, the most effective training technique is to start with easy behaviors, to reinforce those behaviors, and then to shape gradually more complex behaviors.

Applied Behavior Analysis / Behavior Modification

In one way or another, people are almost constantly trying to influence other people’s behavior. Psychologists have developed procedures based on operant conditioning for strengthening their influence for various purposes.

In **applied behavior analysis**, also known as **behavior modification**, a psychologist first determines the reinforcers that are sustaining an unwanted behavior and then tries to alter that behavior by reducing the reinforcements for the unwanted behavior and providing suitable reinforcers for more acceptable behaviors. For example, consider the use of safety belts in cars. Many adults still do not habitually wear safety belts, even though the use of such belts would prevent an estimated 50% of the fatalities and serious injuries in automobile accidents. Many states impose fines for failing to wear seat belts, but such laws are only moderately effective.

One alternative is to reinforce people for wearing safety belts. Virginia Polytechnic Institute invited people parking on campus to sign cards pledging to wear safety belts. All those signing the pledge became eligible for drawings for prizes. Drivers were also invited to affix a display in their car windows, indicating that they wear seat belts. Campus police randomly distributed prize coupons to cars containing that display. This procedure increased the percentage of drivers wearing safety belts from its previous level of 53 to 72% (Thyer & Geller, 1990).

Similarly, applied behavior analysis is used in therapy for psychologically disturbed people, to teach techniques of effective parenting, to encourage medical patients to follow the doctor’s instructions, and in business and industry. In each case, the purpose is to provide clear and immediate reinforcements for the desired behaviors.

CONCEPT CHECK

14. Of the procedures characterized in Table 6.2, which one applies to laws that penalize people for not wearing seat belts? Which one applies to giving prizes to people who do wear seat belts? (Check your answers on page 218.)

Breaking Bad Habits

Some people learn to conquer their own bad habits by means of reinforcements. Nathan Azrin and Robert Nunn (1973) recommend this three-step method:

1. Become more aware of your bad habit. Interrupt the behavior and isolate it from the chain of normal activities. Then, you might imagine an association between the behavior and something repulsive (Yates, 1985). For example, to break a nail-biting habit, imagine your fingernails covered with sewage.
2. If no one else will reinforce you for making progress, provide your own reinforcements. For example, buy yourself a special treat after you have abandoned your bad habit for a certain period of time.
3. Do something incompatible with the offending habit. For example, if you have a nervous habit of hunching up your shoulders, practice depressing your shoulders.

Figure 6.17 shows an example of a college student setting up a list of reinforcements and punishments to support his goal of decreasing his smoking. If he successfully limited his smoking, he would treat himself to a movie. If he

Date: January 1, 1999

Goal: To cut down on my smoking

What I will do: For the first month I will smoke no more than one cigarette per hour. I will not smoke immediately after meals. I will not smoke in bed. In February I will cut back to one every other hour.

What others will do: My roommate Joe will keep track of how many cigarettes I smoke by counting cigarettes in the pack each night. He will keep records of any cigarettes I smoke after meals or in bed.

Rewards if contract is kept: I will treat myself to a movie every week if I stick to the contract.

Consequences if contract is broken: If I break the contract, I have to clean the room by myself on the weekend.

Signatures:

Steve Self
Joe Roommate

FIGURE 6.17 Sometimes, people try to change their own behavior by setting up a system of reinforcements and punishments.

exceeded the limit he had set for himself, he would have to clean the room by himself on the weekend, and he would not go to a movie. Many people set up similar patterns of reinforcement and punishment for themselves, generally without a written contract. If you decide to try this approach, set clear goals, choose realistic reinforcers, and keep track of your successes and failures.

THE MESSAGE

Operant Conditioning and Human Behavior

Suppose one of your instructors announced that everyone in the class would receive the same grade at the end of the course, regardless of performance on tests and papers. Would you study hard in that course? Probably not. Or suppose your employer said that all raises and promotions would be made at random, with no regard to how well you do your job. Would you continue working as hard as possible? Not likely. Do not take it as an insult that some aspects of your behavior resembles that of a pigeon or a rat in a Skinner box. For any organism to survive, its behavior must increase or decrease depending on its consequences. That is the main point of operant conditioning.

S U M M A R Y

- ★ **Reinforcement.** Edward Thorndike introduced the concept of reinforcement. A reinforcement increases the probability that the preceding response will be repeated. (page 205)
- ★ **Operant conditioning.** Operant conditioning is the process of controlling the rate of a behavior through its consequences. (page 206)
- ★ **Extinction.** In operant conditioning, a response becomes extinguished if it is no longer followed by reinforcement. (page 206)
- ★ **Shaping.** Shaping is a technique for training subjects to perform difficult acts by reinforcing them for successive approximations to the desired behavior. (page 207)
- ★ **Reinforcement and punishment.** Behaviors can be reinforced (strengthened) by presenting favorable events or by omitting unfavorable events. Behaviors can be punished (suppressed) by presenting unfavorable events or by omitting favorable events. (page 210)
- ★ **The nature of reinforcement.** The opportunity to engage in a more probable behavior will reinforce a less probable behavior. Something that an individual can exchange for a reinforcer becomes a reinforcer itself. (page 212)
- ★ **Learning what leads to what.** Animals (and people) learn which reinforcement is associated with which behavior. The frequency with which they repeat a given behavior depends on their motivation to receive the reinforcement at the moment. (page 213)

★ **Schedules of reinforcement.** The frequency and timing of a response depends on the schedule of reinforcement. In a ratio schedule of reinforcement, an individual is given reinforcement after a fixed or variable number of responses. In an interval schedule of reinforcement, an individual is given reinforcement after a fixed or variable period of time. (page 213)

★ **Applications.** People have applied operant conditioning to animal training, persuasion, applied behavior analysis, and the breaking of bad habits. (page 215)

Suggestions for Further Reading

- Skinner, B. F. (1953). *Science and human behavior*. New York: Macmillan. A systematic statement of Skinner's positions, written for students.
- Staddon, J. (1993). *Behaviorism*. London: Duckworth. A critique of both the strengths and weaknesses of Skinner's views.

Terms

- reinforcement** an event that increases the future probability of the most recent response (page 206)
- law of effect** Thorndike's theory that a response followed by favorable consequences becomes more probable and a response followed by unfavorable consequences becomes less probable (page 206)
- operant conditioning** or **instrumental conditioning** the process of changing behavior by following a response with reinforcement (page 206)
- visceral responses** activities of the internal organs (page 206)
- skeletal responses** movements of the muscles that move the limbs, trunk, and head (page 206)
- extinction** in operant conditioning, the weakening of a response after a period without reinforcement (page 206)
- stimulus generalization** in operant conditioning, the tendency to make a similar response to a stimulus that resembles one that has already been associated with reinforcement (page 207)
- discrimination** in operant conditioning, learning to respond in one way to one stimulus and in a different way to another stimulus (page 207)
- discriminative stimulus** stimulus that indicates on which occasion a response will produce a certain consequence (page 207)
- belongingness** the concept that certain stimuli are readily associated with each other and that certain responses are readily associated with certain outcomes (page 207)
- shaping** a technique for establishing a new response by reinforcing successive approximations (page 208)
- chaining** a procedure for developing a sequence of behaviors in which the reinforcement for one response is the opportunity to engage in the next response (page 209)
- punishment** an event that decreases the probability that a response will be repeated (page 210)
- positive reinforcement** strengthening a behavior through the presentation of an event such as food (page 210)
- passive avoidance learning** learning to avoid an outcome such as shock by being passive—that is, by inhibiting a response that would lead to the outcome (page 210)

- omission training** learning to suppress a behavior that would lead to the omission of an event such as food (page 210)
- escape learning** learning to escape from an event such as shock (page 211)
- active avoidance learning** learning to make a response to avoid an event such as shock (page 211)
- Premack principle** the principle that the opportunity to engage in a frequent behavior will reinforce a less frequent behavior (page 212)
- disequilibrium principle** the principle that an opportunity to engage in any deprived activity will be a reinforcer because it restores equilibrium (page 212)
- unconditioned reinforcer** an event that is reinforcing because of its own properties (page 212)
- conditioned reinforcer** an event that becomes reinforcing because it has previously been associated with an unconditioned reinforcer (page 212)
- continuous reinforcement** reinforcement for every correct response (page 213)
- intermittent reinforcement** reinforcement for some responses and not for others (page 213)
- schedule of reinforcement** a rule or procedure linking the pattern of responses to the reinforcements (page 213)
- fixed-ratio schedule** a rule for delivering reinforcement only after the subject has made a specific number of correct responses (page 213)
- variable-ratio schedule** a rule for delivering reinforcement after varying numbers of correct responses (page 213)
- fixed-interval schedule** a rule for delivering reinforcement for the first response that the subject makes after a specified period of time has passed (page 214)
- variable-interval schedule** a rule for delivering reinforcement after varying amounts of time (page 214)
- applied behavior analysis** or **behavior modification** a procedure for determining the reinforcers that sustain an unwanted behavior and then reducing the reinforcements for the unwanted behavior and providing suitable reinforcers for more acceptable behaviors (page 216)

10. Begin by determining how this person spends his or her time—for example, exercising, reading, watching television, visiting with friends. Then determine something that he or she has recently not had much opportunity to do. Activities for which one has only limited opportunities become good reinforcers. (page 212)
11. **a.** variable ratio. (You will be reinforced for about one-fourth of your entries to the theater but on an irregular basis.) **b.** variable interval. (The messages appear at unpredictable times, so your responses are reinforced at unpredictable intervals.) **c.** fixed interval. (page 214)
12. In a variable-ratio schedule, the number of responses matters but the timing does not. If you have already checked the stars tonight and found no comets, checking three more times tonight will probably be fruitless. Checking at a later date gives you a better chance. (page 214)
13. The habitual gambler will continue longer, because he or she has a history of being reinforced for gambling on a variable-ratio schedule, which retards extinction. (For the same reason, an alcoholic who has had both good experiences and bad experiences while drunk is likely to keep on drinking even after several consecutive bad experiences.) (page 215)
14. If you think of “not wearing seat belts” as a behavior, then penalizing people for that behavior is a punishment. If (more reasonably) you think of wearing seat belts as the behavior to be changed, wearing seat belts is a way of avoiding punishment, and the example is therefore one of avoidance behavior or negative reinforcement. Giving prizes to people who agree to wear seat belts is an example of positive reinforcement. (page 216)

Web Resources

Positive Reinforcement

server.bmod.athabasca.ca/html/prtut/reinpair.htm

Lyle K. Grant of Athabasca University helps students understand what does and what does not constitute positive reinforcement. Be sure you understand the examples before you begin the practice exercise.

Dr. P's Dog Training

www.uwsp.edu/acad/psych/dog/dog.htm

Mark Plonsky of the University of Wisconsin discusses obedience competition, K9 training, assistance dogs, and working dogs, and offers lots of general information.

Animal Training at Sea World

www.seaworld.org/animal_training/atcontents.html

Sea World trainers use operant conditioning principles with performing animals. The discussion covers primary and conditioned reinforcers, shaping, observational learning, and other principles.

Operant Conditioning

snycorva.cortland.edu/~ANDERSMD/OPER/operant.HTML

This site offers a basic introduction to the key concepts involved in operant conditioning.

Answers to Concept Checks

7. You would need to know whether the bell was always followed by food (classical conditioning) or whether food was presented only if the animal sat up on its hind legs (operant conditioning). (page 206)
8. **a.** positive reinforcement; **b.** punishment or passive avoidance; **c.** escape learning or negative reinforcement; **d.** punishment or passive avoidance; **e.** omission training or negative punishment; **f.** omission training or negative punishment; **g.** avoidance learning or negative reinforcement. (page 211)
9. To be effective, punishments must be quick and consistent. Punishments for drug dealing are neither quick nor consistent. Furthermore, punishment most effectively suppresses a response when the individual has alternative responses that can gain reinforcements. Many people who gain enormous profits by selling drugs would have no alternative legal way to gain similar profits. (page 212)

Other Kinds of Learning

What kinds of learning do not fit neatly into the categories of classical or operant conditioning? What are the special features of these kinds of learning?

How do we learn from the successes and failures of others without trying every response ourselves?

Operant and classical conditioning both depend on an association between a first event (response or CS) and a second event (reinforcement or UCS). The main difference is that, in operant conditioning, the first event is something the individual does, whereas in classical conditioning, the first event is a stimulus presented from the outside. It might seem that all possible examples of learning would fall into either of these two categories; several, however, are difficult to classify or else require another category altogether.

Conditioned Taste Aversions

First, suppose an experimenter repeatedly presents a light followed by a shock. After a few pairings, the animal in the experiment shows an increased heart rate whenever it sees that light. Which kind of learning is this?

Second, suppose the experimenter turns on the light and then waits to see what the animal does. If (and only if) it looks toward the light, it gets a shock. Before long, the animal consistently turns away from the light. Which kind of learning is this?

The first is classical conditioning; the second is operant. But now consider a third experiment: The experimenter turns on a light. The animal looks toward it. Then it gets a shock. Now suppose *on the very next trial* and from then on, the animal avoids the light. What kind of learning took place? It's hard to say. Did the light itself predict the shock (as in classical conditioning) or did the animal's response to the light produce the shock (as in operant conditioning)? When learning occurs reliably after just one pairing, it is difficult to distinguish between classical and operant conditioning.

One kind of learning that does occur reliably after a single trial is an *association between eating something and getting sick*, which we call **conditioned taste aversion**, first documented by John Garcia and his colleagues. This kind of learning shows some special features, including strong learning despite delays of minutes or hours. For example, imagine an experiment conducted using a rat. The same process occurs in other species, including humans, but most of the research is with rats. The experiment begins with a rat drinking a saccharin solution. The effect is strongest with an unfamiliar taste, so let's suppose the rat has never tasted saccharin before. Saccharin tastes sweet, and in small amounts it is neither healthful nor harmful. After the rat has drunk the solution for a few minutes, the experimenter removes the bottle, waits minutes or even hours, and then injects a small amount of poison, thus making the rat moderately ill. The experimenter then waits a day or longer to let the rat recover and then offers it a choice between the saccharin solution and unflavored water. The rat will strongly prefer the unflavored water (Garcia, Ervin, & Koelling, 1966). In contrast, rats that had not been poisoned, or rats poisoned after drinking something else, strongly prefer the saccharin solution to plain water. Evidently, the first group of rats had learned a connection between taste and illness, in spite of the long delay and their having experienced the pairing only once. In many other cases of either classical or operant conditioning, learning is greatest with a 1- or 2-second delay between the events to be associated and hard to demonstrate at all with delays over 20 seconds (Kimble, 1961).

An animal that learns a conditioned taste aversion to a particular food treats that food as if it were foul-tasting (Garcia, 1990). Some ranchers in the western United States have used this type of learning to deter coyotes from eating sheep. They offer the coyotes sheep meat containing low levels of lithium salts or similar poisons. Afterward, as shown in Figure 6.18, the coyotes no longer attack sheep and act as if sheep meat tasted bad. This technique has the advantage of protecting the ranchers' sheep without killing the coyotes, which are a threatened species.

Conditioned taste aversions are special in another regard as well: Recall that an animal can associate something it ate with getting ill hours later. No doubt the animal had many other experiences between its meal and the illness, so why didn't it associate the illness with something else instead? The answer is that animals are predisposed to associate illness mostly with what they eat. In one classic experiment (Garcia & Koelling, 1966), rats were allowed to drink saccharin-flavored water from tubes that were set up so that, whenever the rats licked the water, a bright light flashed and a loud noise sounded. Some of the rats were exposed to X rays (which can induce nausea) while they drank. Others were given electric shocks to their feet 2 seconds after they had begun to drink. After the training was complete, each rat was tested separately with a tube of saccharin-flavored water and a tube of unflavored water that produced lights and noises. (Figure 6.19 illustrates this experiment.)



FIGURE 6.18 This coyote previously fell ill after eating sheep meat containing a mild dose of lithium salts. Now it reacts toward both live and dead sheep as it would toward bad-tasting food.

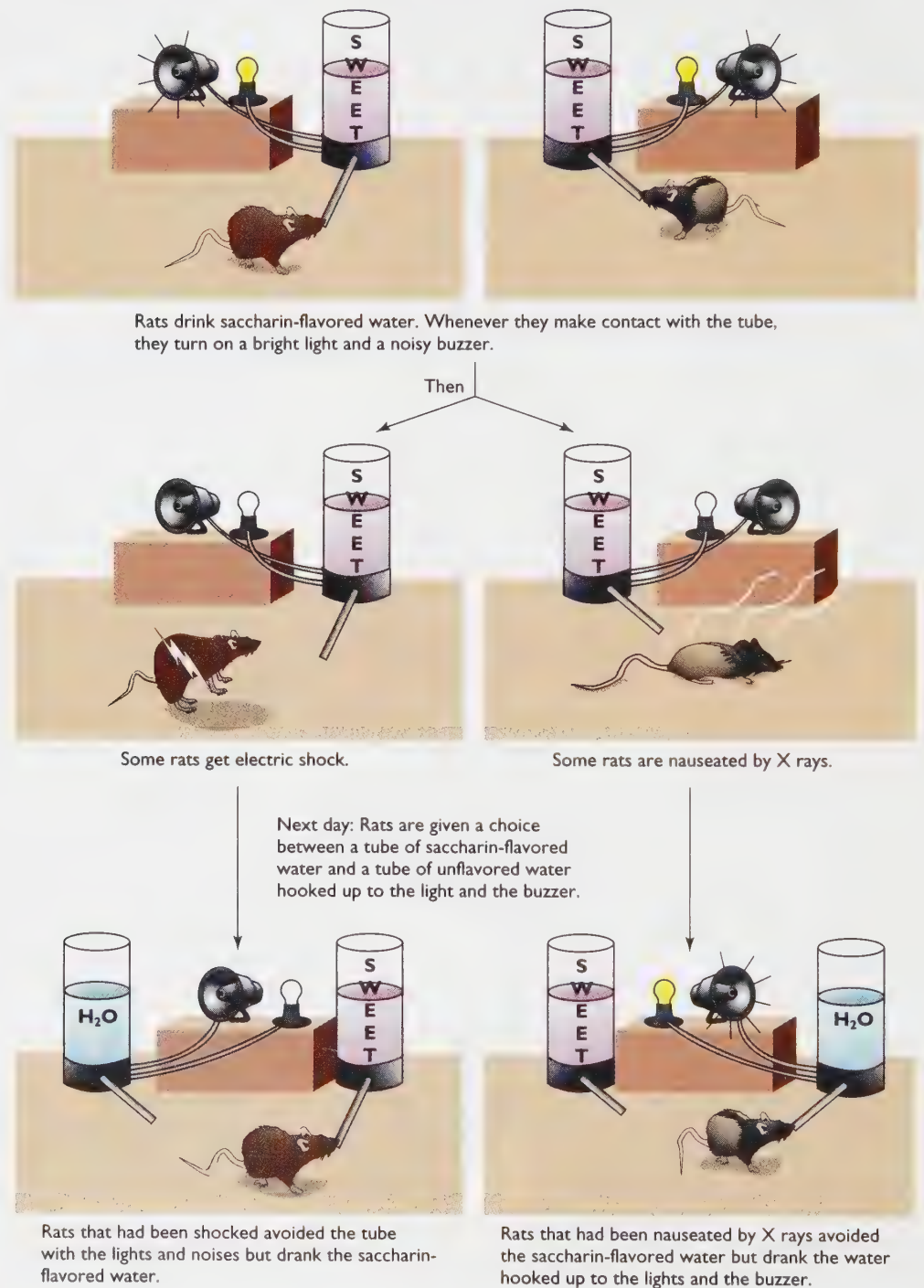


FIGURE 6.19 An experiment by Garcia and Koelling (1966): Rats “blame” tastes for their illness; lights and sounds, for their pain.



A male white-crowned sparrow learns his song in the first months of life, but does not begin to sing it until the next year.

The rats that had been exposed to X rays avoided only the flavored water. The rats that had received shocks while drinking avoided only the tube that produced lights and noises. Evidently, rats (and other species) have a built-in predisposition to associate illness mostly with what they have eaten or drunk and to associate skin pain mostly with what they have seen or heard. (The predisposition to associate illness with food is an example of preparedness, mentioned earlier in this chapter.) Such predispositions are presumably beneficial because foods are more likely to cause internal events, and lights and sounds are more likely to signal external events.

CONCEPT CHECK

15. Name the unusual features of conditioned taste aversions. (Check your answer on page 225.)

Birdsong Learning

Birdsongs brighten the day for people who hear them, but for birds it is earnest business. For most species, the song is limited to males, mostly during spring, the mating season. As a rule, a song indicates, “Here I am. I am a male of _____ species. If you’re a female of my species, please come closer. If you’re a male of my species and you can hear me, you’re too close. If I find you, I’ll attack.” (Among the delights of birdsongs are the exceptions to the rule. Mockingbirds copy all the songs they hear and defend their territory against intruders of any species—sometimes even squirrels, cats, people, automobiles. . . . And Carolina

wrens sing male-and-female duets throughout the year. But on to more relevant matters.)

If you were to rear an infant songbird in isolation from others of its species, would it develop a normal song on its own? Maybe or maybe not, depending on its species. Some species have to learn their song. For example, in several sparrow species, a male will develop a normal song only if he hears the song of his own species. He *learns most readily* during a **sensitive period** *early in life*. The duration of that period varies among species, but generally ends by fall, when the adult males stop singing. The sensitive period lasts longer if the infant hears a live male, such as his father, under natural conditions and ends sooner if he is hearing just a tape recording of the song in a laboratory experiment (Baptista & Petrinovich, 1984; Marler & Peters, 1987, 1988). But if the infant hears the song of another species, he will not imitate it. Evidently, a fledgling sparrow is equipped with mechanisms to produce approximately the right song and ways of identifying which songs to imitate and which to disregard (Marler, 1997).

Song learning differs from the other kinds of learning we have examined. During the sensitive period, the infant bird just listens. We cannot call the song he hears an unconditioned stimulus, because it elicits no apparent response. At no time in this sensitive period does the bird receive any apparent reinforcement. Nevertheless, he forms a representation of how his song should sound, and he begins to practice singing when he reaches sexual maturity the following spring. At that point, we can recognize a trial-and-error process with a strange form of operant conditioning. At first, his song is a disorganized mixture of sounds, somewhat like a babbling human infant. As time passes, he eliminates some sounds and rearranges others until he matches the songs he heard the previous summer (Marler & Peters, 1981, 1982). Once he establishes a song, he can continue singing it even if he becomes deaf.

The point is that the principles of learning vary from one situation to another. If a situation poses special problems—such as food selection, song learning in birds, probably language learning in humans—we can expect to find that species have evolved their own special ways of learning (Rozin & Kalat, 1971).

CONCEPT CHECK

16. What aspects of birdsong learning set it apart from classical and operant conditioning? (Check your answers on page 225.)

Social Learning

According to the **social-learning approach** (Bandura, 1977, 1986), *we learn about many behaviors before we try them the first time. Much learning, especially in humans,*



According to the social-learning approach, we learn many behaviors by observing what others do, imitating behaviors that are reinforced, and avoiding behaviors that are punished. This girl is being blessed by the temple elephant. Others who are watching may later imitate her example.

results from observing the behaviors of others and from imagining the consequences of our own behavior. For example, if you want to learn how to swim, paint pictures, play bridge, or drive a car, you *could* try to learn strictly by trial and error, but you would probably start by watching someone who already knows how to do the activity. When you eventually try it yourself, your attempt will be subject to reinforcement and punishment and therefore falls into the general realm of operant conditioning, but your attempt will be facilitated by your observations of others. It is therefore useful to examine social learning as a special case.

Although psychologists frequently speak of “social-learning theory,” it is not a theory in the sense described in Chapter 2. The social-learning approach is closer to a point of view or a field of emphasis. It focuses on the effects of observation, imitation, setting goals, and self-reinforcement. In this sense, much of human behavior depends on social learning; after all, most school learning is an attempt to learn from the experiences of other people.

Modeling and Imitation

If you visit another country, especially one with customs very different from your own, you may find yourself bewildered about things you used to take for granted. Ordering food in a restaurant is not done the way you did it back home; paying the bill is done differently, too. A hand gesture



such as  is considered friendly in some countries but rude or vulgar in others. Americans visiting Japan often have trouble understanding how to use the baths and toilets. With effort, you can learn foreign customs. Someone may explain them to you, or you may simply watch and copy. You either *model* your behavior after that of others or you



A child will mimic an adult's behavior even when neither one is reinforced for the behavior. This girl attacks a doll after seeing a film of a woman hitting it. People who witness violent behavior, including violence at home, may be more prone than others to turn to violent behavior themselves.

imitate others. On a smaller scale, you follow the same processes to learn the customs of a religious organization, a fraternity or sorority, or a new place of employment.

Albert Bandura, Dorothea Ross, and Sheila Ross (1963) studied the role of imitation for learning aggressive behavior. They asked two groups of children to watch films in which an adult or a cartoon character violently attacked an inflated “Bobo” doll. They had another group watch a film in which the characters did not attack the doll. They then left the children in a room with a Bobo doll. The children who had watched films showing attacks on the doll (and only those children) attacked the doll vigorously, using many of the same movements they had just seen. The clear implication is that children copy the aggressive behavior they have seen in others.

Is the same true for adolescents and adults? This issue warrants our concern, because many popular movies include so much violence. As you read in Chapter 2, the available evidence does not demonstrate that watching violence on television or in movies necessarily causes violent behavior. However, some individuals are more highly influenced than others; some viewers (especially adolescents with a history of violent behavior) identify strongly with a highly violent character in a film. Others may be highly influenced by a film because it resembles an event they have witnessed in their own lives. Many cases have been reported in which people have reenacted scenes they had just seen in a film (Snyder, 1991).

CONCEPT CHECK

17. Many people complain that they cannot find much difference between the two major political parties in the United States, because so many American politicians campaign using similar styles and take similar stands on the issues. Explain this observation in terms of social learning. (Check your answer on page 225.)

Vicarious Reinforcement and Punishment

Six months ago, your best friend quit a job with Consolidated Generic Products in order to open a restaurant. Now

you are considering quitting your own job with Consolidated Generic and opening your own restaurant. How do you decide whether to take this step?

Perhaps the first thing you do is find out how successful your friend has been. You do not automatically imitate the behavior of someone else, even someone you admire. Rather, you imitate behavior that has proven reinforcing for that person. In other words, you learn by **vicarious reinforcement** or **punishment**—that is, by *substituting someone else's experience for your own*.

When a new business venture succeeds, other companies try to figure out the reasons for that success and then try to follow the same course. When a venture fails, other companies try to learn the reasons for that failure and try to avoid making the same mistakes. When a football team wins consistently, other teams copy its style of play. And when a television program wins high ratings, other producers are sure to present look-alikes the following year.

SOMETHING TO THINK ABOUT

Might vicarious learning lead to a certain monotony of behavior and contribute to the lack of variety in the television programs and movies that are offered to the public? How can we learn vicariously without becoming just like everyone else? ★

In many cases, vicarious punishment seems to affect behavior less than vicarious reinforcement does. We are bombarded by reminders that failure to wear seat belts will lead to injury or death, and yet many of us fail to buckle up. Despite widespread publicity about the consequences of driving while intoxicated, using addictive drugs, or engaging in unsafe sex, many people ignore the dangers. Even the death penalty, an extreme example of vicarious punishment, has little demonstrable effect on the murder rate.

Why does vicarious punishment so frequently produce such weak effects? One explanation is that, to be influenced, we must identify with the person who is receiving a vicarious reinforcement or punishment. Most of us think of ourselves as successful people; we see someone who is getting punished as a “loser” and “not like us.” We can therefore continue to ignore the dangers.

The Role of Self-Efficacy in Social Learning

People are selective when choosing who to imitate. We imitate people we regard as successful, people we would like to resemble. Advertisers, keenly aware of this tendency, try to feature endorsements from people consumers are likely to admire and imitate. Cereal and candy advertisements feature happy, healthy children; soft-drink ads feature attractive young adults; ads for luxury cars feature people who look wealthy.

So, when you watch an Olympic diver win a gold medal for a superb display of physical control, do you then go out

and try to make some spectacular dives into a pool? The person is successful, you would like to identify with that person, and if you do, you have just experienced vicarious reinforcement for the diver's success. And yet you probably do not try to imitate the dives. Why not?

If you are like most people, the reason is that you doubt you are capable of duplicating the diver's performance. People imitate someone else's behavior only if they have a sense of **self-efficacy**—*the perception that they themselves could perform the task successfully*. You observe your past successes and failures, compare yourself to the successful person, and estimate your chance of success.

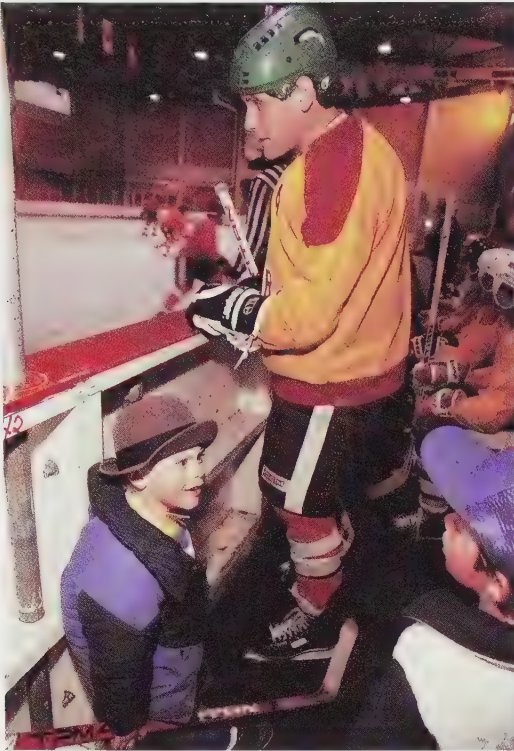
People's sense of self-efficacy generally correlates well with their persistence at a task. For example, people who believe they can succeed at losing weight or quitting smoking or giving up alcohol have a much better chance of prolonged success than do people who doubt their own ability to succeed (Curry, Marlatt, & Gordon, 1987). Of course, as with other correlations, we cannot draw conclusions about cause and effect. Perhaps their sense of self-efficacy helps people to break bad habits; just as likely, people may develop that sense of self-efficacy because they know they can break their bad habits.

Self-Reinforcement and Self-Punishment in Social Learning

We learn by observing others who are doing what we would like to do. If our sense of self-efficacy is strong enough, we try to imitate their behavior. But actually succeeding often requires prolonged efforts. People typically set a goal for themselves and monitor their progress toward that goal. They even provide reinforcement or punishment for themselves, just as if they were training someone else. They say to themselves, “If I finish this math assignment on time, I'll treat myself to a movie and a new magazine. If I don't finish on time, I'll make myself clean the stove and the sink.”



We tend to imitate the actions of successful people, but only if we feel self-efficacy, a belief that we could perform the task well.



We acquire a sense of self-efficacy mostly through our own successes but also partly by watching and identifying with role models.

(Self-punishments are usually pretty mild and seldom actually imposed.)

People who have never learned to use self-reinforcement can be taught to do so. Donald Meichenbaum and Joseph Goodman (1971) worked with a group of elementary-school children who acted impulsively, blurted out answers, and failed to consider the consequences of their actions. To encourage them to set appropriate goals for themselves and to practice self-reinforcement in achieving them, Meichenbaum and Goodman taught the children to talk to themselves while working on a task. For example, a child might say, "Okay, what do I have to do? You want me to copy the picture. . . . Okay, draw the line down, down, good; then to the right, that's it; now down some more and to the left. Good, I'm doing fine so far. Remember, go slowly. Now back up again. No, I was supposed to go down. That's okay. Just erase the line carefully. . . ." After only four training sessions, the children had learned to pause before answering questions, and they were answering more questions correctly.

Unfortunately, self-reinforcement and self-punishment do not always succeed. One psychologist, Ron Ash (1986), tried to teach himself to stop smoking by means of punishment. He decided to smoke only while he was reading *Psychological Bulletin* and other highly respected but tedious publications. By associating smoking with boredom, he hoped to eliminate his desire to smoke. Two months later he was smoking as much as ever, but he was starting to *enjoy* reading *Psychological Bulletin*!

THE MESSAGE

Why We Do What We Do

Almost everything you have done today was a learned behavior—from getting dressed and combing your hair this morning to reading this chapter right now. In fact, you probably would have trouble listing many things you have done today that were not learned. Even your bad habits are examples of learning.

One point that I hope has emerged in this chapter is that learning takes many forms. Classically conditioned salivation, operantly conditioned movements, conditioned taste aversions, and socially learned behaviors occur under diverse circumstances. The underlying mechanisms in the brain may or may not be the same, but at a descriptive level these types of learning differ in some important ways. In short, your behavior is subject to a wide variety of learned influences.

S U M M A R Y

- ★ **Conditioned taste aversions.** Animals, including people, learn to avoid foods, especially unfamiliar ones, if they become ill afterward. This type of learning occurs reliably after a single pairing, even with a delay of hours between the food and the illness. Animals are predisposed to associate illness with what they eat or drink, not with other events. (page 219)
- ★ **Birdsong learning.** Infant birds of some species must hear their song during a sensitive period in the first few months of life, if they are to develop a fully normal song the following spring. During the early learning, the bird makes no apparent response and receives no apparent reinforcement. (page 221)
- ★ **Learning by observation.** We learn much by observing what other people do and what consequences they experience. (page 221)
- ★ **What we imitate.** We tend to imitate behaviors that have led to reinforcement for other people. We are less consistent in avoiding behaviors that have led to punishment. (page 222)
- ★ **Whom we imitate.** We are more likely to imitate the actions of people we admire and people with whom we identify. (page 223)
- ★ **Self-efficacy.** Whether we decide to imitate a behavior that has led to reinforcement for others depends on whether we believe we are capable of duplicating that behavior. (page 223)
- ★ **Self-reinforcement and self-punishment.** Once people have decided to try to imitate a certain behavior, they set goals for themselves and may even provide their own reinforcements and punishments. (page 223)

Suggestion for Further Reading

Bandura, A. (1986). *Social foundations of thought and action*. Englewood Cliffs, NJ: Prentice-Hall. A review of social learning by its most influential investigator.

Terms

- conditioned taste aversion** the tendency to avoid eating a substance that has been followed by illness when it was eaten in the past (page 219)
- sensitive period** a time early in life during which some kind of learning occurs most readily (page 221)
- social-learning approach** the view that people learn by observing and imitating the behavior of others and by imagining the consequences of their own behavior (page 221)
- vicarious reinforcement** or **vicarious punishment** observed reinforcement or punishment experienced by someone else (page 223)
- self-efficacy** the perception of one's own ability to perform a task successfully (page 223)

Answers to Concept Checks

15. Conditioned taste aversions develop after a single trial, despite delays of minutes or hours between food and illness—a longer delay than any that could produce rapid learning in any other situation that psychologists have studied. Also, animals are predisposed to associate foods and not other events with illnesses. (page 221)
16. The most distinctive feature is that birdsong learning occurs during a time when the learner makes no apparent response and receives no apparent reinforcement. Also, at least in certain sparrow species, birdsong learning occurs most readily during an early sensitive period, and the bird is capable of learning its own species' song but not the song of another species. (page 221)
17. One reason that most American politicians run similar campaigns and take similar stands is that they all tend to copy the same models—candidates who have won elections in the past. Another reason is that they all pay attention to the same public-opinion polls. (page 221)

Web Resources

Albert Bandura

www.ship.edu/~cgboeree/bandura.html

- C. George Boeree of Shippensburg University provides a short biography of Albert Bandura, describes early research in social learning, and defines many key terms and concepts.

Memory



7

★ MODULE 7.1

Types of Memory

Why No One Experiment Characterizes All of Memory

Ebbinghaus's Pioneering Studies of Memory /

Differences Among Individuals Because of Interference /

The Importance of Meaningfulness / The Importance of

Distinctiveness / Dependence of Memory on the Method of Testing

The Information-Processing View of Memory

The Sensory Store / Short-Term Memory Versus Long-

Term Memory / The Transfer from Short-Term Memory to Long-Term Memory

A Modified Theory: Working Memory

Other Memory Distinctions

★ MODULE 7.2

Memory Improvement

The Influence of Emotional Arousal

Meaningful Storage and Levels of Processing

Self-Monitoring of Understanding

The Effects of Spacing Out Study Sessions

The Use of Special Coding Strategies

Encoding Specificity / Mnemonic Devices

★ MODULE 7.3

Memory Loss

"Normal" Forgetting

Amnesia After Brain Damage

The Case of H. M., a Man with Hippocampal Damage /

Frontal-Lobe Amnesia / Implicit Memory in Amnesic

Patients / The Implications of Brain-Damage Amnesia

Infant Amnesia

Amnesia of Old Age

★ MODULE 7.4

The Reconstruction of Memory

Reconstruction of the Past

Reconstruction and Inference in List Memory /

Reconstruction of Stories / Hindsight Bias / Eyewitness Identifications

The "False Memory" Controversy

Repression / SUGGESTIONS AND FALSE MEMORIES: WHAT'S

THE EVIDENCE? / Children as Eyewitnesses

Suppose I offer you—for a price—an opportunity to do absolutely anything you want to do for one day. You will not be limited by any of the usual constraints on what is possible. You can travel in a flash from one place to another, visiting as many places as you care to crowd into that single day. You can even travel into outer space, searching for life on other planets. You can travel forward and backward through time, finding out what the future holds in store and witnessing the great events of history, or even prehistoric times. (But you will not be able to alter history.) Anything you want to do—just name it and it is yours. Furthermore, I guarantee your safety: No matter where you choose to go or what you choose to do, you will not get hurt.

Now, how much would you be willing to pay for this once-in-a-lifetime opportunity? Oh, yes, I should mention . . . there is one catch. When the day is over, you will forget everything that happened. You will never be able to recover any memory of that day. Any notes or photos will vanish. And anyone else who takes part in your special day will forget it, too.

Now, how much would you be willing to pay? Much less than before, I am sure; perhaps nothing. Living without remembering is hardly living at all: Our memories are almost the same as our “selves.”



Kutbidin Atamkulov travels from one Central Asian village to another singing from memory the tale of the Kirghiz hero, Manas. The song, which lasts 3 hours, has been passed from master to student for centuries. Human memory can hold an amazing amount of information.

Types of Memory

Do we have different types of memory?

Why do we remember some types of material better than others?

John Horton Conway, a Princeton University mathematics professor, is an accomplished mathematician, magician, game theorist, and computer programmer who is also known for his impressive memory. For example, he has memorized the names of all the stars visible in the northern hemisphere. As he told one interviewer (Seife, 1994, p. 13): “I like knowing things. When a constellation is covered up by a cloud, I predict where the stars will be when the cloud moves away. I really get a feeling of power—knowing what will happen is almost like making it happen.” He can also recite poetry that he learned as a child but has not thought about since then. He memorized π to 1,000 decimal places and persuaded his wife to learn it too. They take long, ro-



Professor Conway is a brilliant mathematician with an excellent memory for most matters, but he has trouble remembering his colleagues' names.

matic walks during which he recites a few places of π , then she says a few, then he says a few. . . .

So, Conway has an outstanding memory, right? Yes, except that during the 20 years he spent at Cambridge University, he never did learn the names of all his colleagues in the math department (Seife, 1994). Examples such as this illustrate a point that psychologists took decades to learn: It is possible to have an excellent memory for some kinds of material but only an average or even a poor memory for other kinds of material, depending on your interest in the material, how and under what circumstances you learned it, and so forth.

Why No One Experiment Characterizes All of Memory

If you have an important test on Wednesday morning, how much do you study on Tuesday night? Some professors advise their students against last-minute cramming, and students occasionally misinterpret that advice to mean that last-minute *reviewing* is useless. (I met one student who proudly told me that he didn't study at all the day before a test. He seemed puzzled by his mediocre grades.) What professors mean is that you should study as you go, not wait until the last night to start. But even if you take this advice, a last-minute refresher does help.

A few years ago someone wrote a proposal for a book called “How to Study in College”; the author offered the opposite advice: Students should postpone their studying until the last possible moment, because if they studied earlier, they would forget it all. If you have ever waited until the night before a test to start your study, you *know* this manuscript was offering bad advice. So how could its author, a college professor who had read at least some research on memory, have come to such a faulty conclusion? And what evidence do we have to refute it?

Ebbinghaus's Pioneering Studies of Memory

The German psychologist Hermann Ebbinghaus (1850–1909) founded the experimental study of memory. Previous researchers had asked people to describe their memories, but they could not measure the accuracy of these reports. Ebbinghaus conducted his research by teaching new material and then measuring memory for that material after various delays. To make sure that the material to be memorized would be unfamiliar, Ebbinghaus invented nonsense syllables, meaningless three-letter combinations such as GAK or VUB. He wrote out 2,300 such syllables, arranged them in random lists (Figure 7.1), and then set out to determine how well people could memorize and retain such lists. He had no cooperative introductory psychology students to draw from

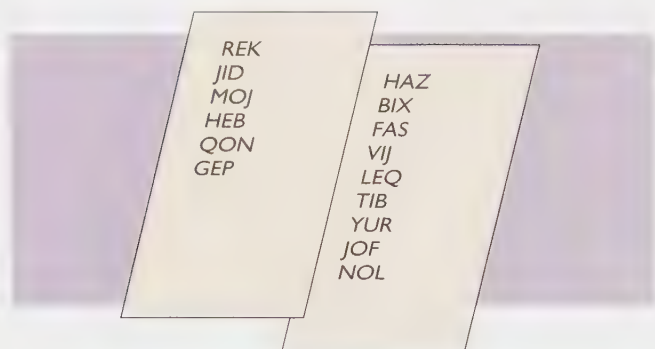


FIGURE 7.1 Hermann Ebbinghaus pioneered the scientific study of memory by observing his own capacity for memorizing lists of nonsense syllables.

for his study, nor friends who would volunteer to memorize lists of nonsense syllables, so he ran all the tests on himself. Over a period of about 6 years, he memorized thousands of lists of nonsense syllables. (He was either very dedicated to his science or uncommonly tolerant of boredom.)

In one experiment, Ebbinghaus memorized several lists of 13 nonsense syllables each and then tested his memory after various delays. (The results appear in Figure 7.2.) He forgot a mean of more than half of each list after one hour and still more after 24 hours. If you take these results seriously, you will decide to do your studying as close as possible to the time of the test. After all, any earlier study is likely to be forgotten.

Of course, then you should also conclude that education itself is pointless. If you will forget most of what you learn within 24 hours, how much will you still remember by the time you graduate, much less a few years after graduation? So, unless the whole idea of education is fundamentally flawed, Ebbinghaus's results must not apply to all memory.

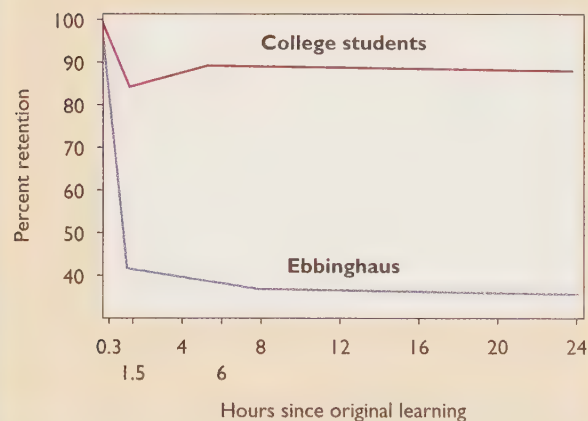


FIGURE 7.2 Blue line: Recall of lists of syllables by Ebbinghaus (1913) after delays of various lengths. Red line: Recall of lists of words by college students after delays of various lengths (based on Koppenaal, 1963). Ebbinghaus learned as fast as other people but forgot faster.

Differences Among Individuals Because of Interference

Suppose we repeat Ebbinghaus's experiment, but instead of doing all the studies on one person, we persuade a large number of college students to learn a list of nonsense syllables, testing some of them after short delays and some after long delays. The top line of Figure 7.2 shows the results of one such study: Even after 24 hours, most students still remember most of the nonsense syllables on the list (Koppenaal, 1963).

Why do you suppose most college students can remember a list so much better than Ebbinghaus did? You may be tempted to say that college students did so well because they are so intelligent. True, no doubt, but Ebbinghaus was no dummy either. Or you might suggest that college students have had "so much practice at memorizing nonsense." (Sorry if you think that's true.) But Ebbinghaus had memorized a lot of nonsense himself. In fact, the problem was that poor Hermann Ebbinghaus had memorized too much nonsense—literally thousands of lists of syllables (Figure 7.3). When anyone memorizes large amounts of similar material, the memory becomes a bit like a cluttered room: Something seems to be lost just because it is buried among other items and hard to find. Learning vast amounts of information does not block new learning (any more than a cluttered room prevents you from bringing in another piece of clutter), but it increases the risk of confusing old items with new ones and therefore forgetting.



FIGURE 7.3 Ebbinghaus could learn new lists of nonsense syllables, but he forgot them quickly because of interference from all the previous lists he had learned. People who memorize many similar lists start to confuse them with one another.

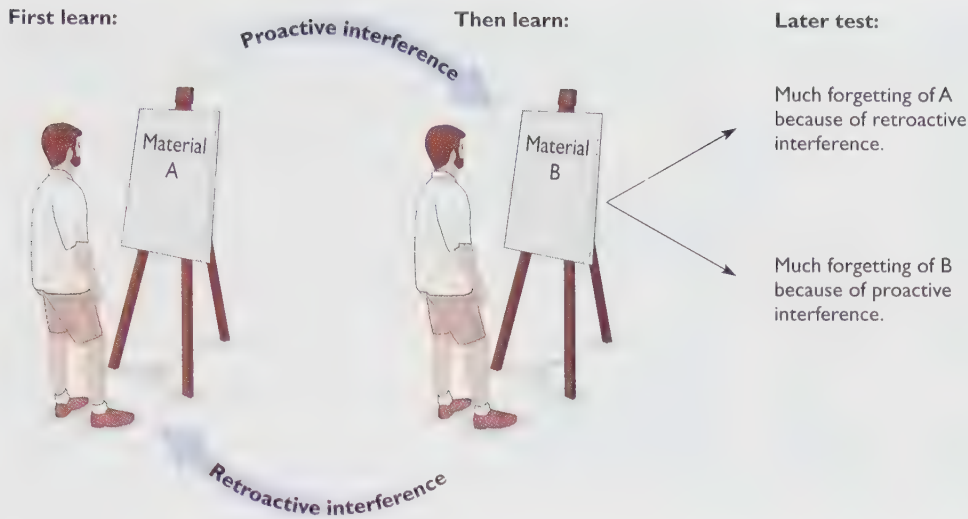


FIGURE 7.4 When someone learns two similar sets of materials, each interferes with the other. The old interferes with the new by proactive interference; the new interferes with the old by retroactive interference.

If you learn several sets of related materials, the old interferes with the new and the new interferes with the old. The old materials increase forgetting of the new materials through proactive interference (acting forward in time); the new materials increase forgetting of the old materials through retroactive interference (acting backward in time).

Figure 7.4 shows the difference between these two kinds of interference. Interference is responsible for a great deal of our everyday forgetting. You may forget where you parked your car at the local shopping mall, because of proactive interference from all the previous times you parked in the same lot. You may forget the French vocabulary list that you studied three weeks ago, because of retroactive interference from other lists you have studied since then.

The application of this idea to Ebbinghaus is simple: Because he had memorized so many lists of syllables, he had massive proactive interference and he forgot new lists much faster than other people would. Therefore, the results he collected on himself are very misleading if they are applied to anyone else. The moral of the story: When you want to memorize something, beware of studying anything else very similar to it. (Studying unrelated material poses no problem.)

CONCEPT CHECKS

1. Professor Tryhard learns the names of his students every semester. After several years, he learns them as quickly as ever but forgets them faster. Does he forget because of retroactive interference or proactive interference?
2. Remember the concept of spontaneous recovery from Chapter 6, page 198? Can you explain it in terms of proactive interference? (What is learned first? What is learned second? What would happen if the first interfered with the second?) (Check your answers on page 240.)

The Importance of Meaningfulness

Ebbinghaus, you will recall, was memorizing nonsense syllables. As you might guess, people tend to remember meaningful material better than they remember nonsense.

For example, in one experiment, two groups of people examined either picture (a) or picture (b) in Figure 7.5 and listened to the paragraph in part (c) (Bransford & Johnson, 1972, p. 131). Note that the paragraph makes sense if you have seen picture A but not if you have seen only picture B. As you might expect, the people who had seen picture A remembered about twice as much of the paragraph as those who had seen only picture (b).

How well do you suppose you remember the meaningful, important events of your life? Marigold Linton (1982) wrote notes about at least two important personal events each day for 6 years and recorded the date on the back of each note. At various times, she drew notes at random from the pile and tested her ability to recall the approximate date of each event. She found that she remembered the dates of about 95% of events 1 year old and a slightly lower percentage of older events. Similarly, Willem Wagenaar (1986) wrote index cards recording 2,400 personal events over 6 years. At the end of the 6 years, he tested himself by reading part of each card (for example, "what happened") and trying to recall the rest (for example, who, where, and when). He was able to recall at least a little information about almost every event. In short, we generally remember meaningful events well, even years later.

The Importance of Distinctiveness

TRY IT YOURSELF

We also tend to remember distinctive or unusual material. Read the following list and then recall immediately, in any order, as many items as you can:

potato, asparagus, cauliflower, turnip, broccoli, Egypt, beans, corn, peas, rutabaga, squash, cabbage



a



b

If the balloons popped, the sound would not be able to carry since everything would be too far away from the correct floor. A closed window would also prevent the sound from carrying since most buildings tend to be well insulated. Since the whole operation depends on a steady flow of electricity, a break in the middle of the wire would also cause problems. Of course, the fellow could shout, but the human voice is not loud enough to carry that far. An additional problem is that a string could break on the instrument. Then there could be no accompaniment to the message. It is clear that the best situation would involve less distance. Then there would be fewer potential problems. With face to face contact, the least number of things could go wrong.

c

FIGURE 7.5 In an experiment by Bransford and Johnson (1972), one group of people looked at picture (a) and another group looked at picture (b). Then both groups heard the paragraph in part (c). The paragraph is meaningful only if you have seen picture (a).

One of the items you are most likely to recall is *Egypt*, because it is different from the others: It is the only word beginning with a capital letter and the only word that refers to anything other than a vegetable.

Similarly, we tend to remember unusual people or people with unusual names. If you meet several men of rather ordinary appearance and similar names, like John Stevens, Steve Johnson, and Joe Stevenson, it may take you a long time to get their names straight. You will be quicker to remember a 7-foot-tall, redheaded man named Stinky Rockefeller. The tendency to remember unusual items better than more common items is known as the **von Restorff effect**, after the psychologist who first demonstrated it (von Restorff, 1933).

Dependence of Memory on the Method of Testing

Ebbinghaus tested his memory by requiring himself to repeat the correct syllables in the correct order. Might his method have underestimated his memory? For example, most of us occasionally find ourselves unable to remember someone's name, or the formula for the volume of a sphere, or another fact that we once knew. Still, we know we haven't forgotten it altogether; later, a reminder may bring back the memory.

How well someone appears to remember something depends on how a psychologist tests the memory. The simplest way is to ask for **recall**. To recall something is to produce it, as you do on essay tests or short-answer tests. For instance, if I ask you, "Please name all the children in your fourth-grade class," you will probably have trouble doing so, partly because you confuse the names of the children in your fourth-grade class with those you knew in other grades. (Remember the influence of proactive and retroactive interference.)

You will remember more with **cued recall**, a method in which you receive significant hints about the material.

For example, I might show you a photograph of all the children in your fourth-grade class, or else I might give you a list of their initials (Figure 7.6). Try this: Cover the right side of Table 7.1 with a piece of paper and try to identify the authors of each book on the left. (This is the recall method.) Then uncover the right side, revealing each author's initials, and try again. (This is cued recall.)

With **recognition**, a third method of testing memory, a person is asked to identify the correct item from several choices. People can usually recognize more items than they can recall. For example, I might give you a list of 60 names and ask you to check off the correct names of children in your fourth-grade class. Multiple-choice tests use the recognition method to test memory.



FIGURE 7.6 Can you recall the names of the students in your fourth-grade class? Trying to remember without any hints is *recall*. Using a photo or a list of initials is *cued recall*. If you tried to choose the correct names from a list, you would be engaged in *recognition*. If you compared how fast you relearned the correct names and how fast you learned another list, you would be using the *savings* (or *relearning*) method.

TABLE 7.1 The Difference Between Recall and Cued Recall

Instructions: First try to identify the author of each book listed in the left column while covering the right column (recall method). Then expose the right column, which gives each author's initials, and try again (cued recall).

BOOK	AUTHOR
<i>Moby Dick</i>	H. M.
<i>Emma</i> and <i>Pride and Prejudice</i>	J. A.
Hercule Poirot stories	A. C.
Sherlock Holmes stories	A. C. D.
<i>I Know Why the Caged Bird Sings</i>	M. A.
<i>War and Peace</i>	L. T.
This book	J. K.
<i>Canterbury Tales</i>	G. C.
<i>The Origin of Species</i>	C. D.
<i>Gone with the Wind</i>	M. M.
<i>The Color Purple</i>	A. W.
<i>Les Misérables</i>	V. H.

(For answers, see page 240, Answer A.)

A fourth method, the **savings, or relearning, method**, can detect weaker memories than the other methods will. Suppose you cannot name some of the children in your fourth-grade class (recall method) and cannot pick out their names from a list of choices (recognition method). If I presented you with the correct list of names, you might learn it faster than you would learn an unfamiliar list of names. You relearn something more quickly than you learn something new. In other words, you save time when you relearn material that you learned in the past. The amount of time saved (time needed for original learning minus the time for relearning) is a measure of memory.

An overall conclusion: We revere Ebbinghaus as the pioneer who started scientific memory research. However, many memories are remembered far longer and better than Ebbinghaus's results implied. His memory was impaired by proactive interference and by dealing with meaningless and nondistinctive information.

CONCEPT CHECK

3. Each of the following is an example of one method of testing memory. Identify each method.
 - a. Although you thought you had completely forgotten your high-school French, you do much better in your college French course than your roommate, who never had French in high school.
 - b. You don't have a telephone directory and are trying to remember the phone number of the local pizza parlor.

- c. After witnessing a robbery, you have trouble describing the thief. The police show you several photographs and ask whether any of them was the robber.
- d. Your friend asks, "What's the name of our chemistry lab instructor? I think it's Julie or Judy something." (Check your answers on page 240.)

The Information-Processing View of Memory

One important theme of memory research since about 1950 has been the search for distinctions among types of memory. According to one view, the **information-processing model** of memory, human memory is analogous to the memory system of a computer: Information enters the system, is processed and coded in various ways, and is then stored (Figure 7.7). According to a version of this model that was long considered the standard, information first enters temporary storage (as when information is typed into a computer) and later enters permanent storage (as when information is entered onto a disk). Still later, in response to a cue from the environment, a person can recover the information from permanent storage (Atkinson & Shiffrin, 1968). According to this model, we distinguish three types of memory: a very brief sensory store, short-term memory, and long-term memory. We shall examine this model and its strengths and weaknesses and then consider alternative views.

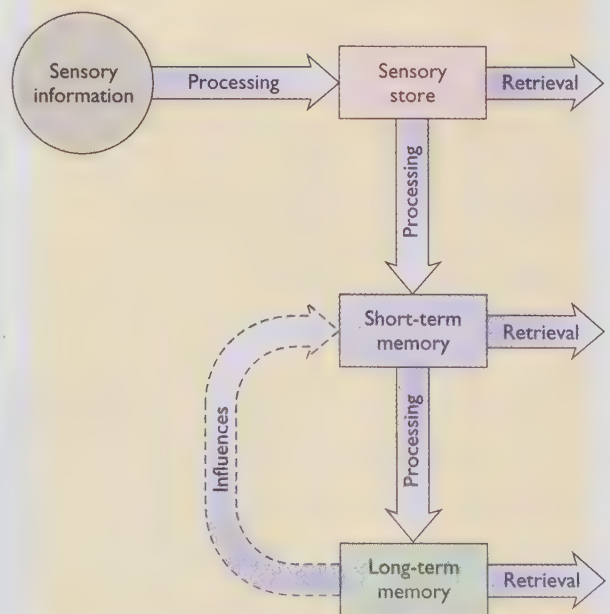


FIGURE 7.7 The information-processing model of memory resembles a computer's memory system, including temporary and permanent memory.



A bolt of lightning flashes through the sky for just a split second, but you can visualize it in detail for a short time afterward. That image is held momentarily in your sensory store.

The Sensory Store

Every memory begins as an exposure to a sensory stimulus. After you see or hear something for even a split second, you can temporarily recall minor details about it, because it has entered the sensory store, a very brief storage of sensory information. However, unless you immediately and actively attend to this information, it will fade rapidly as new information replaces it in the sensory store.

George Sperling (1960) tested how much information people can retain in the sensory store. He flashed an array like the one shown in Figure 7.8 onto a screen for 50 milliseconds. When he asked viewers to report as much of the whole array as they could, he found that they could recall a mean of only about four items. If he had stopped his experiment at that point, he might have concluded that viewers stored only a small fraction of an array.

But Sperling knew that it takes several seconds for a person to report even a few items, and he knew that memories in the sensory store would probably fade during that time. To test that possibility, he told viewers he would ask them to report only one row of the array, but he did not tell them which row. After flashing the array on the screen, he immediately used a high, medium, or low tone to signal which row the viewers were to recall. Most people could name all of the items in whichever row he indicated. Evidently, nearly all of the information in the array was briefly available to them.

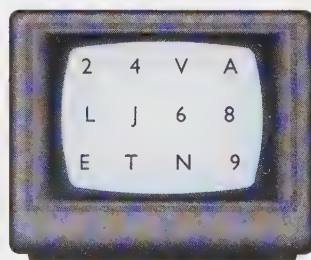
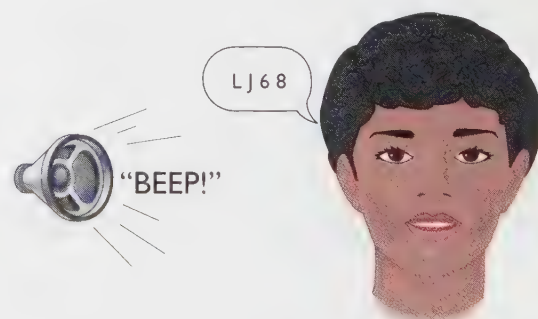


FIGURE 7.8 George Sperling (1960) flashed arrays like this on a screen for 50 milliseconds. After the display went off, a signal told the viewer which row to recite.

Stimulus array flashed on screen for 0.05 seconds



Within the next 0.1 to 0.3 seconds, high, medium, or low tone signals indicate which row to say

Participant says the correct row

When he waited for even one second before signaling which row to recall, though, viewers were less likely to recall that row. That is, for information in the sensory store, the rule is “use it or lose it,” and you had better use it fast.

Is this phenomenon really memory, or is it perception? It is a borderline case. Not everything falls neatly into our human-made categories.

SOMETHING TO THINK ABOUT

Sperling demonstrated the capacity of the sensory store for visual information. How could you demonstrate the capacity of the sensory store for *auditory* information? ★

Short-Term Memory Versus Long-Term Memory

Of all the vast information that passes through our sensory store, a small amount catches our attention enough to get stored and processed a little longer. But even for information that catches our attention, some gets stored permanently and some does not. According to the traditional version of information-processing theory, we should distinguish between short-term memory, temporary storage of the information that someone has just experienced, and long-term memory, a relatively permanent store of mostly meaningful information. Figure 7.9 compares the sensory store and short- and long-term memory. Short-term and long-term memory differ in their dependence on retrieval cues, their capacity, and their tendency to decay.

Dependence on Retrieval Cues If you try to recall a telephone number that you read just seconds ago, presumably stored in your short-term memory, either you can recall it easily or it is already gone. You don’t need reminders or hints, and chances are that no reminder or hint would do you much good anyway. In contrast, to get information from long-term memory, you need a retrieval cue, an association that elicits the memory. Some retrieval cues work better than others. For example, the retrieval cue “the city you were born in” should enable you to retrieve the name of one

city from your long-term memory. The retrieval cue “U.S. city whose name is also the name of an animal” probably does not enable you to retrieve anything quickly. You laboriously go through all the cities (or animals) you can think of until eventually you come up with an answer—most likely, Buffalo, New York. (Conceivably, you might also think of Caribou, Maine; White Pigeon, Michigan; Anaconda, Montana; or Deadhorse, Alaska.) Retrieving long-term memories is sometimes a difficult, effortful task.

Differences in Capacity The capacity of long-term memory is so vast that we cannot easily measure it. Unlike computers, people never fill their memories so full that they have no room to store something new. (Of course, also unlike computers, people are constantly dumping some of their memories.) Short-term memory, in contrast, has a relatively small, easily measured capacity. Read each of the following sequences of letters; then look away and try to repeat them from memory. Or read each aloud and ask a friend to repeat it.

**TRY IT
YOUR-
SELF**

EHGPH
JROZNQ
SRBWRCN
MPDIWFBS
ZYPPIAFMO
BOJFKFLTRC
XUGJDPFSVCL

Most normal adults can repeat a list of approximately seven items—letters, in this case. Some people can remember eight or nine; others, only five or six. George Miller (1956) referred to the short-term memory capacity as “the

magical number seven, plus or minus two.” When people try to repeat a longer list, however, they may fail to remember even the first seven items. It is somewhat like trying to hold several objects in one hand: You can hold a certain number, depending on their size, but if you try to hold too many you will drop them all (Figure 7.10). However, when we refer to a limit of about seven “items,” the size of an item can vary. You can store more information if you can organize it into familiar **chunks, or meaningful units.** (See Figure 7.11.) As an illustration, examine each of these sequences of numbers and then try to repeat them:

106614921776

31416–271828–1414–1732

Could you do it? The first list was either easy or impossibly long, depending on whether you recognized the sequence as three historical dates (the Norman invasion of England in 1066, Columbus’s arrival in America in 1492, and the Declaration of Independence in 1776). The second number is already separated into four chunks, but the separation will help you only if you recognize the chunks as the approximate values of π , 3.1416 . . . ; e , 2.71828 . . . ; the square root of two, 1.414 . . . ; and the square root of three, 1.732.

With practice, people can learn to recognize larger and larger chunks of numbers. One student from Carnegie-Mellon University (Ericsson, Chase, & Falcon, 1980) volunteered for an experiment on memorizing digits. At the beginning, he could repeat only about seven digits at one time. But over the course of a year and a half, working 3 to 5 hours per week, he gradually improved until he could repeat

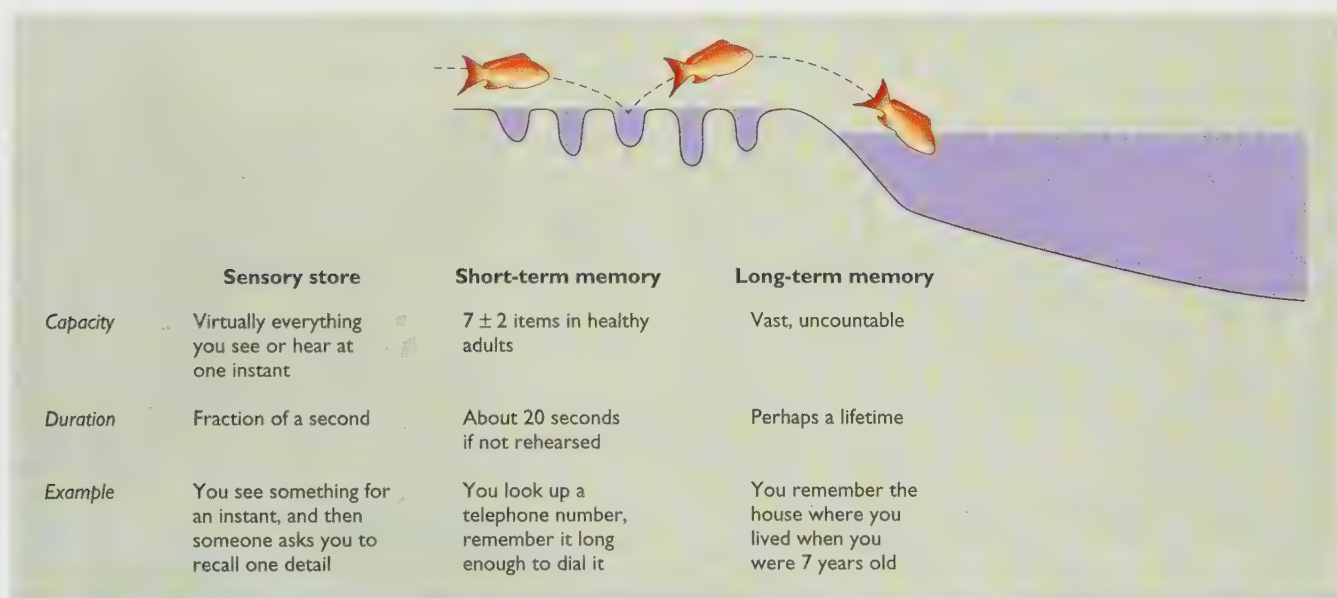


FIGURE 7.9 After about 1 second, you can no longer recall information from the sensory store. Short-term memories can be recalled up to about 20 seconds without rehearsal—much longer if you continually rehearse them. Long-term memories decline somewhat, especially at first, but you may be able to retrieve them for a lifetime. Your address from years ago is probably in your long-term memory and will continue to be for the rest of your life.